

AUTOMATIC WATER LEVEL CONTROLLER FOR SUMP/SUBMERSIBLE PUMP



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The normal sump pump water level controller cannot be used for submersible pumps, as the method to start the motor is different. The submersible pump comes with a starter unit designed for it. To switch it on/off automatically some additional circuits are required. Presented here is an automatic water level controller for

level sensor circuit gets its inputs from probes of three sensors (A, B and C), which are placed in overhead tank (OHT). Depending upon the water level, the level-sensor circuit activates control relay RL1 to switch pump on/off.

The dry sump/no-flow sensors (A' and D) provide the required signal depending upon the water level in the sump, or OHT inflow conditions, and

sensors A (common), B (minimum), C (maximum), transistors (T2, T3, T5 and T7), and control relay RL1. Initially, when the tank is empty, the sensors are not in contact with water, which puts transistors T3 and T7 in cut-off regions. Transistor T2 gets positive bias through resistors R2 and R4, which drives transistor T2 into saturation. Control relay RL1 is energised to switch on the pump motor through its normally open (N/O) contacts. The pump starts pumping water into OHT.

As the water level rises, sensors A and B come in contact with water. A very small amount of DC current (few micro amps) flows through sensors A and B and resistor R6 to provide positive bias to transistor T3. Its emitter is connected to the collector of transistor T7, which is in cut-off state. Transistor T3 also remains in cut-off state. However, transistor T2 still remains in saturation. The pump continues to pump water into the OHT.

When maximum level sensor C comes in contact with water, transistor T7 gets positive bias and provides ground path by driving transistor T7 into saturation. Transistors T3 and T7 go into saturation, which results in the removal of positive bias from the base of transistor T2 driving it into cut-off and de-energising relay RL1, which results in the pump motor to stop. Simultaneously cut-off transistor T2 provides positive bias to transistor T5 through resistor R13 and then transistor T5 starts conducting. As the emitter of transistor T5 is connected to sensor C, transistor T7 continues to get positive bias even when water level falls below sensor C. This latching action holds the pump in off state till the water level falls below the minimum

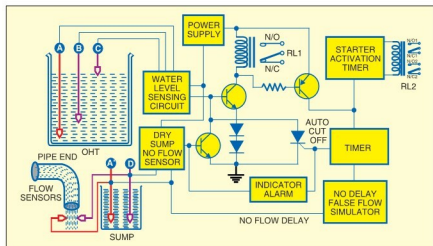


Fig. 1: Block diagram of the automatic water level controller for sump/submersible pump

submersible as well as sump pumps.

The essential features of this controller include:

1. Water level-sensing and control circuit
2. Dry sump/no-flow sensor and delay circuit
3. Auto timer cut-off circuit
4. Submersible pump-starter activation circuit

Circuit description

Fig. 1 shows the block diagram and Fig. 2 shows the circuit diagram of the automatic water level controller for sump/submersible pumps. The water

activate the circuit to enable or disable the control relay driving transistor. The timer circuit produces selectable time delay outputs to activate a silicon control rectifier (SCR) to switch-off the pump controller as desired. The starter activation circuit energises relay RL2 for two to four seconds, which activates the submersible pump starter. Starter activation circuit can be omitted in case the water level controller is used for the sump.

The functioning of individual blocks is as follows:

Water level sensing and control circuit. This circuit consists of level